Stress-Aware Driving App: AI PRD

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Summary

The Stress-Aware Driving App flags sustained driver distress from a live heart signal and responds in escalating stages, from a warning to a simulated pull-over onto the shoulder.

The problem: a driver who is overwhelmed or losing capacity has no automatic safety net, and a heart event at the wheel is rare but severe. Heart disease caused 683,491 US deaths in 2024 (CDC), and about 1.3% of drivers in surveyed crashes were in a medical emergency (NHTSA via SWOV).

Today it is a working prototype built as separate layers: sensor intake, guardrails, an AI scoring layer, an escalation agent, and a dashboard with a freeway simulation. All vehicle control is simulated. A human can cancel every escalation, and the app never calls 911 on its own.

It is not a medical device. Heart rate and heart-rate variability cannot separate stress from a cardiac event, so the product flags distress and never diagnoses.

Goals and non-goals

The goal is to catch sustained distress early enough to reduce risk, without ever acting on a single noisy reading.

Goals

- Begin slowing the simulated car after 5 seconds of sustained distress (current default setting).
- Respond in escalating, reversible stages instead of one hard stop.
- Keep a human in control: the driver can cancel every escalation.
- Make every decision explainable and replayable from logs.
- Keep all vehicle control inside a simulator.

Non-goals

- Diagnosing heart attacks or any other medical condition.
- Controlling a real vehicle, or calling 911 automatically.

- Replacing medical devices or certified driver-monitoring systems.
- Camera-based emotion recognition in the current release.

Principles

Flag, don't diagnose. Fail safe and fail visibly. A human confirms first. Log every decision.

Users and scenarios

The primary user is the driver; three other roles shape what the product must do when the driver cannot act.

Persona	Need	How the product serves them
Driver	Stay safe when overwhelmed, and keep control	Staged response the driver can cancel at any point
Emergency contact	Know when the driver may need help	Notified only if the driver does not respond; must opt in first
Bystander	Act quickly if the driver is unresponsive	Hazards on, shoulder stop, manual Call 911 button
Builder and evaluator	Understand and tune the system	Readable logs, live metrics, evals, adjustable thresholds

The product must handle these scenarios.

Scenario	Expected behavior
Calm drive	Normal state, lane 1, full speed (70 mph in the simulator)
Brief spike	Agent watches only; no action unless it lasts 5 seconds
Sustained distress	Car slows and merges right, lane by lane
Severe distress for 20 seconds	Shoulder stop, hazards on, 15 second countdown
Driver taps "I'm OK"	Escalation cancelled; no new prompt in that episode
No reply to the countdown	Personal contact notified (simulated by default)

Scenario	Expected behavior
Sensor drops out	Visible warning; must never look like a calm driver

Scope

The prototype already covers the full path from heart signal to simulated safe stop; the next release closes its gaps in robustness and validation.

Area	Built today	Next
Signal intake	BLE Heart Rate Service via bleak; labelled demo mode	Phone relay for watches that do not broadcast
Guardrails	Plausibility checks, stale-data fallback, speed rate limiter	Dead-man's alert, confidence gate, noise filter
AI layer	Baseline, plus BPM and HRV distress score	Per-driver baselines; learned model compared with the heuristic
Agent	Normal, watch, intervene, escalate states	Tuning from recorded sessions
Response	Freeway simulator, lane changes, shoulder stop	Reliable notification retries
Notifications	Simulated contact alert; optional SMS (Twilio) and email (SMTP)	Contact opt-in and acknowledgment
Observability	JSONL event log, /metrics, /events	Log purge and privacy controls
Evaluation	Six scenario evals	Replay evals, skin-tone and motion testing, camera cross-check

Out of scope: real-vehicle control, automatic 911 calls, and any diagnosis.

Functional requirements

Twenty requirements define the product: 16 are built and 4 are next, with the dead-man's alert (FR-7) the top priority.

ID	Requirement	Layer	Priority	Status
FR-1	Accept live BPM and RR intervals over the BLE Heart Rate Service	Intake	P0	Built
FR-2	Fall back to a labelled demo mode when no sensor is found	Intake	P1	Built
FR-3	Support watches through a phone relay or broadcast mode	Intake	P1	Next
FR-4	Reject BPM outside 30 to 220 and RR intervals outside 270 to 2,000 ms	Guardrails	P0	Built
FR-5	Treat data older than 8 seconds as stale and show a visible warning	Guardrails	P0	Built
FR-6	Limit speed changes to 0.15 units per 200 ms tick	Guardrails	P0	Built
FR-7	Raise a dead-man's alert when the sensor disconnects or loses skin contact	Guardrails	P0	Next
FR-8	Block intervention until the baseline is set and a minimum number of valid readings exists (threshold to be set)	Guardrails	P1	Next
FR-9	Set a resting baseline from the first 15 seconds	AI layer	P0	Built
FR-10	Score distress 0 to 100: up to 60 points from BPM above baseline, up to 40 from low HRV (RMSSD)	AI layer	P0	Built
FR-11	Move through normal, watch, intervene and escalate on sustained elevation (5 seconds, then 20 seconds at a score of 80 or more)	Agent	P0	Built
FR-12	Reset to normal when the score falls below 60	Agent	P0	Built
FR-13	Slow the simulated car and merge right, lane by lane, when intervening	Response	P0	Built
FR-14	On escalation, stop on the shoulder with hazards on and start a 15 second cancel countdown	Response	P0	Built

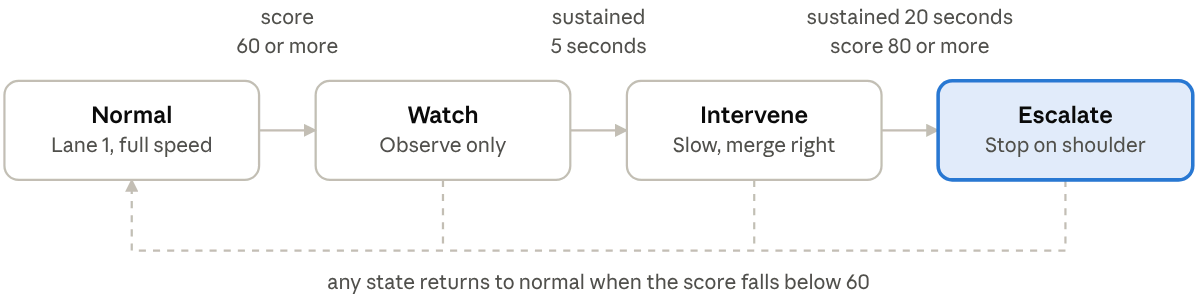
ID	Requirement	Layer	Priority	Status
FR-15	If the driver does not reply, notify one personal contact by SMS or email, never 911	Notifications	P0	Built
FR-16	After a cancel, do not prompt again until the episode ends	Notifications	P0	Built
FR-17	Retry failed notifications and escalate if no one acknowledges	Notifications	P1	Next
FR-18	Always show a manual Call 911 button; never dial automatically	Dashboard	P0	Built
FR-19	Log every assessment, guardrail reject and agent action; expose /metrics and /events	Observability	P0	Built
FR-20	Show gauge, agent state and reasoning, speed, lane, and a manual override for testing	Dashboard	P1	Built

P0 means required for a safe demo; P1 means important but can follow.

AI requirements

The AI layer is a transparent heuristic plus a time-aware agent, not a trained model, so every decision traces back to a few visible inputs.

The agent escalates one step at a time, and any recovery resets it



At escalate: 15 second countdown. The driver can cancel; no reply notifies a personal contact.

agent states · 4 states, 3 escalations, 1 reset

Each step up needs sustained distress; a score below 60 resets the agent from any state.

Item	Definition
Inputs	BPM and RR intervals from one wearable sensor
Derived features	Baseline BPM from the first 15 seconds; BPM elevation over baseline; RMSSD over the last 20 RR intervals
Outputs	Distress score 0 to 100, agent state, target speed, plain-language reasoning
Decision rule	A score of 60 or more starts the timers; 5 seconds sustained intervenes; 20 seconds sustained at 80 or more escalates
Agent actions	Slow down, alert driver, suggest pull-over, log incident, start the emergency countdown
Model type	Rule-based today; a learned model must beat the heuristic on recorded sessions before it replaces it

The values 60, 80, 5 seconds and 20 seconds are starting defaults, not validated clinical thresholds.

Human-in-the-loop rules

- The driver can cancel any escalation during the 15 second countdown.
- Silence is treated as possible incapacity, so it notifies a personal contact, never emergency services.
- Every decision shows its reasoning on screen.
- The agent may act only inside the simulator; it has no interface to a real vehicle.

Evaluation plan

Six scenario evals pass today (6 of 6); five more are needed before any claim about real-world accuracy. Criteria marked proposed are targets, not validated results.

Eval	What it checks	Pass criterion	Status
Calm driver	A resting profile never triggers intervention	No intervention; score under 40	Built

Eval	What it checks	Pass criterion	Status
Sustained stress	Agent reaches intervene on sustained elevation	State is intervene or escalate	Built
Extreme sustained stress	Agent escalates and fires alert and pull-over actions	State is escalate; both actions fired	Built
Recovery	Agent resets when stress drops	State returns to normal	Built
Implausible reading	Guardrail rejects an impossible BPM (300)	Reading rejected	Built
Speed smoothing	Speed never jumps straight to the minimum	First-tick speed stays above the minimum	Built
Replay on recorded sessions	False interventions and missed episodes on real data	No intervention in calm sessions; intervene within 6 seconds of sustained crossing (proposed)	To build
Sensor dropout	Stale data and disconnects raise a visible warning	Warning within 8 seconds in every test (proposed)	To build
Noise and motion	A single outlier or motion artifact never changes agent state	Zero state changes from single outliers (proposed)	To build
Notification flow	Cancel sends nothing; silence sends exactly one message	Zero messages after cancel; one per episode on silence (proposed)	To build
Fairness	Accuracy across skin tones and movement levels against a chest-strap reference	Gap within an agreed tolerance (to be set)	To build

Track five metrics across these evals: false-intervention rate on calm sessions, detection delay after a sustained crossing, missed-episode rate on labelled episodes, time to a sensor-loss alert, and how often drivers cancel.

Safety, risk and privacy

The largest risk is silent failure: today a dead battery or loose strap shows only a status line, so real distress could look like a calm drive.

Risk	Why it matters	Mitigation	Status
Silent sensor loss	Real distress would look calm	Dead-man's alert with a distinct warning (FR-7)	Next
False positive	A needless slowdown surprises the driver	5 second watch period, rate-limited speed, cancel button, simulator only	Built
Mistaken for a diagnosis	Over-trust in a heart-rate signal	Docs say distress only; add the same wording on screen	Partly built
Notification failure or fatigue	The message never arrives, or the contact stops reading	One message per episode today; retries and acknowledgment (FR-17)	Partly built
Contact never agreed	A person is alerted without consent	Contact opts in before being added	Next
Health data exposure	Heart data is sensitive health information	Local-only log today; add purge after a set number of days and an option to stop logging raw BPM	Next
Unequal accuracy	Wrist sensors can read worse with motion, loose straps and skin tone	Fairness eval against a chest-strap reference	Next
Real-vehicle misuse	Software slowing a real car could cause a crash	No actuator interface; real control needs functional-safety validation (ISO 26262) and regulatory approval	Built by design

Heart data stays on the local machine in the current design; nothing is sent to a cloud service except an optional SMS or email to the contact.

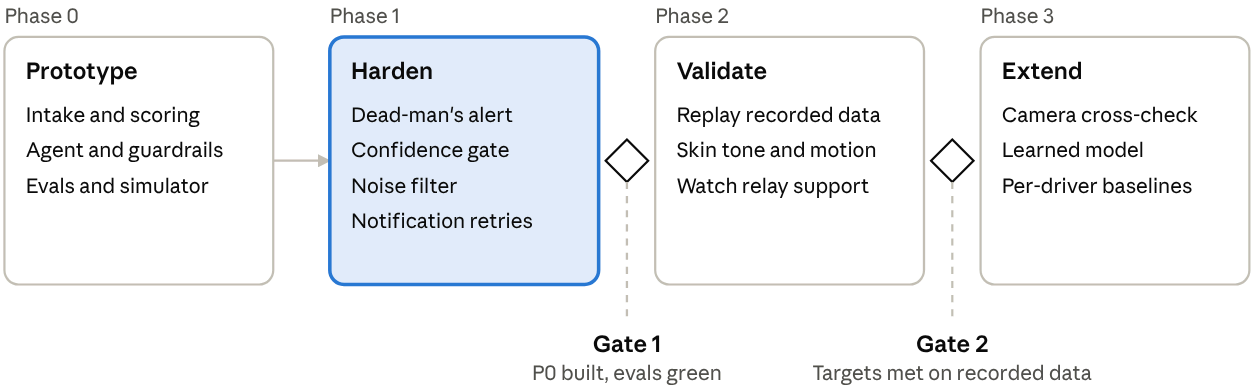
Success metrics and roadmap

This release succeeds when the full path works end to end in the simulator and every P0 requirement is built; accuracy claims wait for recorded data.

North Star: safe-response rate. The share of sustained-distress test episodes where the car starts slowing within 6 seconds, measured on a test set in which calm sessions cause zero false interventions. The metrics below are its inputs. No baseline exists yet; the replay eval must be built first, and the target is to be set once recorded data exists.

Metric	Target	Where measured
Scenario evals passing	100% (6 of 6 today)	Eval suite
P0 requirements built	15 of 15 (14 today; FR-7 outstanding)	Functional requirements table
False interventions in calm replay	0 (proposed)	Replay eval, to build
Sustained crossing to intervention	6 seconds or less (proposed: 5 second timer plus one tick)	Replay eval, to build
Sensor loss to visible alert	8 seconds or less (proposed)	Dropout eval, to build
Notifications sent after a cancel	0	Notification eval, to build

Hardening comes before any new capability



No real-vehicle use without functional-safety validation (ISO 26262) and regulatory approval.

roadmap · 4 phases, 2 gates

Gate 1 needs every P0 requirement built and the evals green; gate 2 needs the targets met on recorded data.

Open questions, dependencies and sources

Five questions need an owner before the hardening phase starts.

Open questions

- Which watch is supported first, given that Fitbit does not expose live heart rate over Bluetooth?
- What distress thresholds suit each driver? The current 60, 80, 5 second and 20 second values are defaults.
- How many valid readings must exist before the agent may intervene (FR-8)?
- How does an emergency contact opt in, and how do they acknowledge an alert?
- Is a camera signal worth adding as a cross-check, given its lighting and occlusion limits?

Dependencies

- A BLE heart-rate sensor, or a watch that can broadcast one.
- Optional Twilio and SMTP accounts for real SMS and email.
- Recorded, labelled sessions for replay evals; none are collected yet.

Evidence base (retrieved October 2026)

Claim	Source
Heart disease caused 683,491 US deaths in 2024, the leading cause	CDC FastStats
About 805,000 US heart attacks a year	CDC Heart Disease Facts
Job strain linked to 23% higher heart attack risk; about 3.4% of heart attacks attributable if causal	ISHN report of the Lancet study
Driver medical emergencies in 1.3% of drivers in surveyed crashes	NHTSA study via SWOV
Most drivers with cardiac arrest at the wheel stopped without injury (30-case autopsy series)	Antecol and Roberts via SWOV
39,254 US traffic deaths in 2024	Trucknews on NHTSA data

The full research worksheet, stress_heart_crash_research.xlsx, holds the wider evidence and its limits.